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Life-history costs make perfect sprouting maladaptive in two herbaceous perennials

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Summary

1. Why some herbaceous plant species refrain from sprouting in some years is a longstanding puzzle in plant ecology. When vegetatively 'dormant', the plant lives as a rootstock, but does not produce or maintain photosynthetic tissue. During this time, energy may be remobilized from resource reserves or acquired from mycorrhizal fungi, although the mechanisms are still poorly understood. If vegetative dormancy is adaptive, it may be in response to a harsh environment, to life-history costs, or as a bet-hedge against unpredictable environmental variability.

2. We tested whether vegetative dormancy is adaptive via game theoretical analysis of deterministic and stochastic ahistorical and historical life-history models parameterized with long-term census data for two long-lived plants, *Anemone americana* and *Cypripedium parviflorum*. The ahistorical deterministic model provided a test of the hypothesis that dormancy is adaptive in response to a generally harsh environment and/or short-term life-history costs, while the historical model tested whether long-term costs drove the evolution of dormancy. The stochastic ahistorical model provided a test of whether dormancy is a bet-hedging trait.

3. We found that vegetative dormancy is an adaptive consequence of life-history costs of growth to survival and that these costs may be operating under a variable but generally harsh environment. Such costs led to sprouting and survival probabilities that generally increased with size in adults but never reached unity and decreased with size in juveniles. Historical deterministic models particularly predicted observed sprouting frequencies, while deterministic ahistorical and stochastic models did not, suggesting although the environment is likely stressful and fluctuates between harsh and mild states, short-term costs and temporal stochasticity alone do not explain observed sprouting frequencies.

4. *Synthesis*. Life-history costs can drive the evolution of seemingly paradoxical traits. In particular, growth can lead to survival costs that may become significant in future years. These costs may be incurred via the use of stored reserves that, once used, cannot be used in the next several years. Such costs are the currency favouring the evolutionary maintenance of vegetative dormancy in two distantly related perennial plant species and may account for dormancy throughout the plant kingdom.

Key-words: *Anemone*, *Cypripedium*, ESS, evolutionary stable strategy, game theory, *Hepatica*, plant development and life-history traits, prolonged dormancy, sprouting frequency, vegetative dormancy

Introduction

> An odd feature of the depletion curves for the orchids is that the number of survivors appears to go up as well as down! Clearly the number of survivors can never increase. The explanation is that the orchids appear to be capable of disappearing from the above-ground population for a year, or perhaps two. . . .It may be that this habit is more common than we know. – John Harper (1977)

In herbaceous plants, sprouting conveys the ability to assimilate energy and to reproduce. Sprouting usually includes the production of photosynthetic tissue, which converts sunlight into a useable and storable form of energy (Hoch, Richter & Korner 2003). Likewise, sprouting results in above-ground support for flowers for reproduction (Wyka 1999). Given these advantages, it is paradoxical that some herbaceous plant species forego sprouting in some growing seasons (Shefferson 2009). How is it advantageous to forego sprouting when doing so prevents both energy production and, in most species, reproduction in that year? J. Harper's famous plant ecology text notes this phenomenon (Harper 1977), but by the mid-1970s vegetative dormancy was still just a misunderstood 'nuisance phenomenon' in plant

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population studies (Shefferson *et al.* 2001), and even today the search for underlying mechanisms and an adaptive context continues (Shefferson 2009; Gremer & Sala 2013; Tuomi *et al.* 2013).

Vegetative dormancy (*aka* prolonged dormancy) is the complete lack of sprouting by an herbaceous plant in one or more growing seasons (Lesica & Steele 1994). It occurs in at least 11 different plant families and 53 species, including ferns, buttercups and orchids (Harper 1977; Shefferson 2009). Vegetatively dormant plants neither produce nor maintain leaves and flowers during the dormant period, so they can neither photosynthesize nor sexually reproduce. Physiologically, stored energy reserves are mobilized during vegetative dormancy (Gremer, Sala & Crone 2010), and in some cases, the mycorrhiza may also contribute carbon-based energy (Bidartondo *et al.* 2004). However, in some cases, what is referred to as vegetative dormancy may also be a misidentified consequence of bud herbivory (Gregg 2011), or an artefact of variability between individuals in the within-year timing of sprouting (Gilbert & Lee 1980).

Vegetative dormancy is a below-ground component of the life cycles of many herbaceous perennial plant species and has historically challenged the development of a common demographic theory for herbaceous plant populations (Tamm 1948). It is often associated with periods of high mortality and so may be a life-history cost (Shefferson *et al.* 2003; Gregg & Kery 2006). However, vegetative dormancy and mortality often correlate negatively with plant size (Shefferson 2006), suggesting that lower mortality may not be a consequence of dormancy but that both mortality and dormancy increase in a harsh environment. Vegetative dormancy increases following defoliation and correlates with climatic variables related to seasonal temperature and annual precipitation (Kéry & Gregg 2004; Shefferson, Kull & Tali 2005), and so may be an adaptation to environmental stress. The survival of long-lived herbaceous plants when dormant is also contingent on previous size (Jäkäläniemi *et al.* 2011), hinting that long-term growth trends of individuals may be important determinants of sprouting frequency. The probability of dormancy increases following reproduction, and so dormancy may sometimes result from reproductive costs (Primack & Stacy 1998). Thus, the ecological context of vegetative dormancy in most species in which it has been found suggests that it has a relationship with fitness and therefore has the potential to be adaptive (Shefferson 2009).

Why should vegetative dormancy be adaptive? First, sprouting may engender costs of being above-ground, which may include greater herbivory and water stress (Shefferson *et al.* 2011). Such costs would be driven by environmental factors and would create correlations between sprouting frequency and some environmental variables. They may also cause higher mortality in sprouting plants than in vegetatively dormant plants in some years. Second, high growth may be intrinsically costly to future growth, survival or reproduction in the short or long term, resulting in sprouting frequencies that are low when a plant is small and young and increase with both age and size. For example, in some species, the probability of flowering is greater following a year of vegetative dormancy than a year of sprouting (Lesica & Crone 2007; Shefferson *et al.* 2011). In this case, vegetative dormancy may act as a recuperation period, particularly if it entails the reaccumulation of spent resources via the mycorrhiza or other means (Shefferson 2009). Third, flowering may be costly (Primack & Stacy 1998), and such costs may be exacerbated if flowering is followed by above-ground growth in the next growing season, particularly if above-ground growth is also sometimes costly. Finally, the environment may vary in quality across time, and if this temporal variability is strong enough, then vegetative dormancy may be favoured to minimize the negative impacts of the bad years on fitness via buffering strategies that appear maladaptive in the short term (Gremer, Crone & Lesica 2012). Here, vegetative dormancy would be a conservative bet-hedging trait, resulting in greater long-term fitness by minimizing the temporal variance in fitness, but also resulting in seemingly lower fitness in the short term by temporarily foregoing high growth and fecundity for a strategy that preserves survival as much as possible (Philippi & Seger 1989; Gremer, Crone & Lesica 2012). For example, if above-ground growth is also costly, then high above-ground growth in a good year may contribute to the mobilization of stored reserves that would otherwise be useful in a stressful year. If such stressful years are particularly harsh and occur unpredictably, then high growth in one year may be followed by high mortality in large plants in the following year, but lower mortality in dormant plants (Shefferson & Roach 2010).

We tested whether natural sprouting frequencies that involve vegetative dormancy reflect intrinsic life-history costs, extrinsic ecological costs of being above-ground, or temporal environmental stochasticity. We used long-term demographic census data of wild populations of two long-lived herbaceous perennial species: the small yellow lady's slipper orchid, *Cypripedium parviflorum* (hereafter *Cypripedium*), and liverleaf, *Anemone americana* (hereafter *Anemone*), to develop mixed models of demographic functions used to parameterize matrix projection models that were tested for an evolutionarily stable strategy (ESS) for sprouting via evolutionary invasion analysis. We hypothesized that vegetative dormancy is an ESS in response to (i) the extrinsic costs of producing or maintaining above-ground tissue, (ii) intrinsic life-history costs, potentially involving growth or reproduction and (iii) stochastic environmental variation across time. We used ahistorical and historical matrix modelling approaches combined with evolutionary invasion analysis to assess the impacts of short- versus long-term life-history costs and environmental stochasticity on optimal sprouting frequency.

Materials and methods

STUDY SPECIES AND SITE

This study involved one population each of the small yellow lady's slipper orchid, *Cypripedium parviflorum* Salisb. var. *parviflorum*

(family Orchidaceae) and liverleaf, *Anemone americana* (DC.) H. Hara (family Ranunculaceae). Both are long-lived (> 30 years) herbaceous perennials of forest edges and interiors with complex life cycles (Fig. 1). The former was located in Gavin Nature Preserve, a 3-ha wet meadow in Lake Villa, Illinois, USA, within a tall grass community dominated by *Andropogon gerardii* and other grasses. The latter was located in Whitehall Forest, a 336-ha mixed deciduous and pine forest plantation in Athens, Georgia, USA, in a shaded understorey shared with other herbaceous perennial species including *Polygonatum* spp.

FIELD METHODS

At Gavin Nature Preserve, we censused all *Cypripedium* sprouts every year in late May/early June from 1994 to 2012 (1157 total plants monitored). We also conducted a fruiting census every July from 2000 to 2012. We did not monitor seedlings because they sprout throughout the growing season, whereas adults all sprout by mid-May. Monitoring involved measurement of location relative to one of several fixed points in the meadow, size (as the number of sprouts), flowers per sprout and fruits per sprout. Sprouts were considered to belong to the same physiological individual if they occurred ≤ 20 cm of each other, which is a criterion that works well due to relatively low population density, most plants having ≤ 4 sprouts and low recruitment rates (Shefferson *et al.* 2001). Via closed population mark–recapture modelling in a repeated census conducted over 2 weeks, we estimated the probabilities of detecting an individual never seen before and of redetecting a previously observed and sprouting individual, as 0.920 ± 0.032 and 0.910 ± 0.035, respectively (Shefferson *et al.* 2001).

At Whitehall Forest, 6 20 × 24 m demography grids were established in 1999, and all plants within them were monitored every year until 2006 (3873 total plants monitored). Unlike *Cypripedium*, *Anemone* plants were easily identifiable rosettes composed of 2.03 ± 0.01 leaves. The study grids were divided into 2 × 2 m cells, and every plant was individually flagged and monitored in annual censuses until 2004. Plants in a subset of 16 cells per grid were monitored until 2006. Two censuses were conducted each year, one in early spring to monitor flowering and seed production and one in late spring to assess seedling emergence (Giladi 2004; Warren 2007).

ANALYTICAL METHODS

Estimating vital rates

We identified all observable instances of vegetative dormancy in the data sets as years in which a plant was not seen, between years in which the plant had been seen. We parameterized population-specific general linear mixed models from the census data to predict the probability of survival between years t and $t + 1$, the probability of sprouting in year $t + 1$, size in year $t + 1$, the probability of flowering in year $t + 1$, the number of flowers in year t, the probability of fruiting in year t and the number of fruits in year t. For each population, we developed both

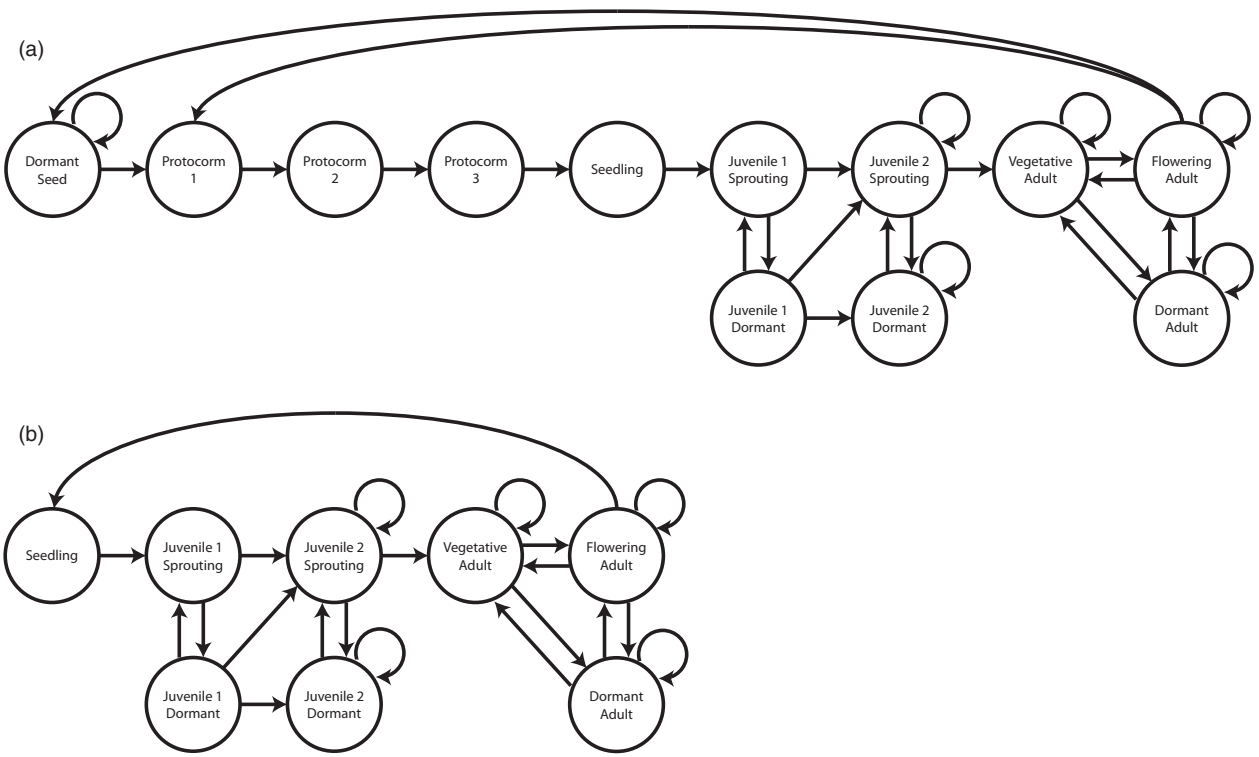

Fig. 1. Life cycle models for *Cypripedium parviflorum* (a) and *Anemone americana* (b). (a) *Cypripedium* begins life as a dust seed, which may germinate the next year or become dormant. It grows into a protocorm, which generally grows for 3 years until a seedling forms. Seedlings develop into juveniles in 1 year, and juveniles resemble small adults but do not flower. At least 2 years of juvenile growth occur prior to the development of the adult vegetative plant. Adults may sprout and flower, sprout without flowering or become dormant. Juvenile and adult sprouting stages are size-stratified, with two juvenile size classes per above-ground juvenile stage, and nine flowering and nine non-flowering size classes in adults. (b) *Anemone* begins life as a seed that either germinates into a seedling in the following year or dies. The next year it will be a juvenile and remain one for at least 2 years prior to becoming an initially non-flowering adult. Adults sprout and flower, sprout without flowering or become dormant. Juvenile and adult sprouting stages are size-stratified, with eight juvenile size classes per above-ground juvenile stage, and 13 flowering and 13 non-flowering size classes in adults.

'ahistorical' mixed models assessing demographic trends between years t and $t + 1$, and 'historical' mixed models assessing trends between years t and $t + 1$ as functions of status in year $t-1$ and year t (Ehrlén 2000). The fixed factors in the most parameterized ahistorical models included size in year t (models of survival, sprouting, growth, flowering probability, and number of flowers), flowering status in year t (models of survival, sprouting, growth, and flowering probability) and the number of flowers in year t (models of probability and number of fruits), as well as all interactions. Historical models included these terms as well as flowering status in year $t-1$ (all models), growth between years $t-1$ and t (all models) and size in year $t-1$ (models of fruiting probability and number of fruits), as well as all fixed effect interactions. Random effects included year and plant identity. Size (measured as number of sprouts) was Poisson-distributed in *Cypripedium* and Gaussian-distributed in *Anemone* (measured as length of basal leaf). Flowering status was binomial, and numbers of flowers and fruits were Poisson-distributed. We developed all biologically relevant simplified versions of these models by systematically removing terms, beginning with fixed effect interactions. The best-fit models were the models with the lowest AIC and were used for matrix building and invasion analysis (Burnham & Anderson 2002).

We also used the ahistorical best-fit models for *Cypripedium* to assess the influence of climate on vital rates. We created models in which the previous year's mean annual temperature, number of days below freezing and total annual precipitation were added as fixed effects to each of the best-fit models. These climatic variables were chosen on the basis of previous analyses showing that they exert some influence on some *Cypripedium* vital rates (Shefferson *et al.* 2001) and because they did not exhibit significant collinearity. We created all reduced models and compared via AIC. The influences of climatic variables on vital rates were inferred via the slopes on climatic terms in the new best-fit models. These models were not used for matrix development and were used only with *Cypripedium* because it was the only data set long enough for meaningful statistical tests of climate variation.

We analysed size trajectories of new recruits in both populations, where new recruits were defined as individuals first observed ≥ 7 years after the start of recording in *Cypripedium* to limit the artefactual inclusion of long-term dormant plants and as seedlings in *Anemone*. Because new recruits were small for ≥ 2 years prior to falling within the size profile of the typical adult, and because new recruits were not capable of flowering, we developed mixed models separately for juveniles and adults.

Our approach may miss some population turnover and vegetative dormancy occurring immediately prior to death and may artefactually suggest that vegetatively dormant individuals do not die. We repeated our analyses using mark–recapture analysis to remove the potential bias resulting from this treatment of survival during dormancy. This approach yielded results consistent with those presented here, but was subject to the loss of temporal resolution in key rates due to parameter redundancy (Schaub *et al.* 2004; Kéry, Gregg & Schaub 2005). We also attempted mixed modelling with the exclusion of the two first years and the two last years in each data set, since those are the years in which misidentification of dormant individuals as either new recruits or dead individuals is most likely, but found no differences from the results presented here.

Population projection modelling

Mixed models were used to develop three life-history function-based Lefkovitch matrix models for each species: ahistorical deterministic, historical deterministic, and ahistorical stochastic. In *Cypripedium*, adults ranged in size from 0 (vegetatively dormant) to 15 sprouts, with most individuals ranging in size from 0 to 9 sprouts in a given year. In *Anemone*, basal leaf sizes ranged from 3 to 75 mm, with dormant plants having no leaves and most sprouting individuals falling below 62 mm in leaf size. We created 10 adult size classes in *Cypripedium* corresponding to each number of sprouts (from 0 to ≥ 9) and 14 adult size classes in *Anemone*, defined by 12 evenly spaced size mesh points between 3 and 62 mm and including dormancy as a separate stage. Adult stages included flowering and non-flowering versions of these size classes, except for vegetative dormancy which was non-flowering (note that size classes are collapsed in Fig. 1). Juvenile size classes included vegetative dormancy and sizes 1 and 2 in *Cypripedium*, and vegetative dormancy and sizes 1–8 in *Anemone*. Including seed dormancy and protocorm stages (*Cypripedium* only), seedlings, juvenile classes and all adult classes, there were 29 and 46 stages for ahistorical models of *Cypripedium* and *Anemone*, respectively.

Matrices included all survival–transition terms and fecundity and were parameterized as products of component demographic functions estimated via mixed models (Fig. S1). In *Anemone*, all transitions were estimable from the data. In *Cypripedium*, seeds, protocorms and seedlings were not tracked, and so we used estimates from the literature for all transitions relevant to these stages (Kull 1999; Nicolè, Brzosko & Till-Bottraud 2005).

Historical matrices were modelled using the two-dimensional approach proposed by Ehrlén (2000). Stages were identified by status over consecutive time steps, with matrix elements defining probabilities and rates of transition from stages representing combinations of status in years $t-1$ and t, to stages representing combinations of status in years t and $t + 1$ (Fig. S2). This led to 395 and 913 stages for *Cypripedium* and *Anemone*, respectively. Deterministic ahistorical and historical matrices were developed for each population factoring out all year effects from mixed models, and annual matrices were developed incorporating year effects for stochastic modelling.

Our matrix modelling approach was similar to the integral projection models that are currently being utilized in population and evolutionary analyses (e.g. Metcalf *et al.* 2008). However, integral projection models use finely spaced mesh points to create high-dimension matrices separately for fecundity and for survival growth. Our use of historical models prevented the use of IPMs due to computational constraints (i.e. the typical ahistorical IPM has matrix dimensionality of $\sim 100 \times 100$, while our historical matrices in *Anemone* alone were almost an order of magnitude larger; a historical IPM matrix for *Anemone* would likely have had dimensions of $\geq 10\,000 \times 10\,000$).

Density dependence

Our evolutionary invasion analyses relied on density dependence in key vital rates to simulate intraspecific competition. To determine which rate to use, we explored the *Cypripedium* and *Anemone* data sets for negative density dependence in key vital rates. We measured density as the number of conspecific individuals per 1 m^2 and 4 m^2 cell in *Cypripedium* and *Anemone*, respectively. In *Cypripedium*, we counted individuals as full, physiological integrated plants with all associated sprouts, rather than as single sprouts, while in *Anemone* all rosettes were treated as individuals. We assessed the impact of density in year t on annual survival probability between years t and $t + 1$ (juveniles and adults separately), sprouting probability in year t (juveniles and adults separately) and flowering probability in year t. In

each case, we developed a mixed model in which the response was a function of density (fixed effect) and year (random effect), using function *glmer* in package *lme4* for *R* 3.0.2 (Bates, Maechler & Bolker 2012; R Core Team 2013). In *Cypripedium*, we also tested density effects with larger grid squares (4 m^2, 25 m^2, and 100 m^2), but found no difference in results from 1 m^2 grid squares.

Evolutionary invasion analysis

We tested whether vegetative dormancy is adaptive via evolutionary invasion analysis. Invasion analysis is a game theoretical method to identify evolutionary stable strategies and is particularly useful for finding optimal values of continuously varying traits and in organisms with complex life histories. First, a range of biologically plausible strategies are identified in the trait of interest. Then, a clonal population is simulated from one founding individual with one of these plausible strategies, referred to as the resident strategy, and is allowed to reach some maximal level via negative density dependence in a vital rate hypothesized to be subject to intraspecific competition (chosen as in the Density dependence section, above). Once the population reaches its maximal level, an invading clonal individual of a different strategy is introduced. This invader acts as a rare mutant in a dense population, and the growth rate of the invading strategy while still rare shows whether it will be driven to extinction by the resident or not. This is repeated with all possible combinations of resident and invader strategies to map a frequency-dependent fitness landscape for the trait, referred to as a pairwise invasibility plot (PIP; Fig. 2). The $y = x$ line identifies the region in which the resident and invader strategies are the same, and the invader should have an instantaneous growth rate of 0 ($r_{inv} = 0$). In cases where this is the only $r_{inv} = 0$ line, evolutionary invasion analysis predicts that the ESS is either the minimal or maximal strategy, depending on whether the invader population grows when adopting a lower or higher level of the trait, respectively (Fig. 2a). If any other $r_{inv} = 0$ lines occur and cross the $y = x$ line, then an intermediate level of the trait corresponding to the intersection is an ESS (Fig. 2b,c).

To test whether vegetative dormancy is adaptive, we first identified our biologically plausible strategies as a range of intrinsic sprouting tendencies, given as deviations to the *y*-intercept in the linear model of sprouting used to develop our Lefkovitch matrices. For example, per the associated best-fit model (Table S1), the linear model of sprouting used in ahistorical, deterministic invasion analysis for *Cypripedium* was

$$\mathrm{Logit}(P_X) = a + d + b_{siz} * siz_t \quad (1)$$

where P_X is the sprouting probability in year $t + 1$ of a plant that had been size X in year t conditional on survival in that interval, a is the *y*-intercept estimated from the best-fit model, d is the altered sprouting strategy (a scalar deviation to alter intrinsic sprouting tendency, with positive values increasing sprouting tendency and decreasing vegetative dormancy, and negative values doing the opposite), b_{siz} is the estimated coefficient for size in year t in the best-fit model and siz_t is size in year t. We then developed baseline matrices using these deviations and conducted population simulations with all possible combinations of resident and invading strategies, using the main vital rate identified as negatively density-dependent in density dependence analysis. In deterministic simulations, the baseline matrix was derived from the best-fit models of vital rates with random year effects cancelled, while in stochastic simulations, the annual matrices served as baseline matrices and were shuffled with resampling across the time span of the simulation. We repeated these analyses using negatively density-dependent recruitment, but found no difference from the results presented here.

We used ESS patterns in these analyses to infer the adaptive nature of vegetative dormancy. We hypothesized that if vegetative dormancy is maladaptive, then all six invasion analyses should yield patterns in which invaders with greater tendencies to sprout than residents should always outcompete them (e.g. Fig. 2a). In contrast, if vegetative dormancy is adaptive, then at least one invasion analysis should yield an intermediate equilibrium point below which invaders with greater sprouting tendencies outcompete residents, but above which invaders with greater sprouting tendencies are outcompeted by residents (e.g. Fig. 2b,c). We bootstrapped the original data set and repeated any analysis in which such an equilibrium point existed 1000 times to estimate the standard error of the equilibrium and asked if it differed

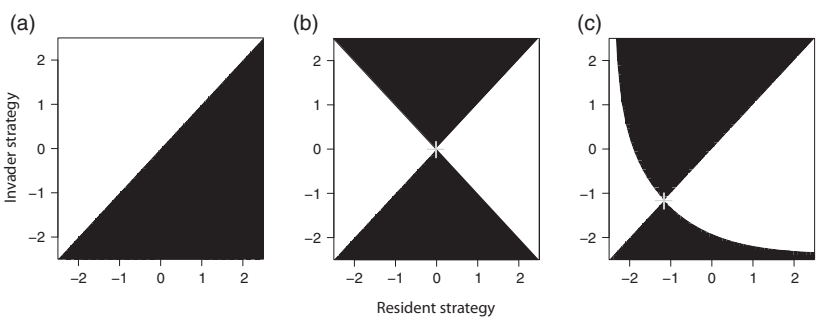

Fig. 2. Example pairwise invasibility plots (PIPs) showing some possible evolutionary outcomes when a rare mutant strategy (the invader) invades a one-strategy population at maximal density (the resident). Shading shows the early fitness of the invader after its introduction, with white zones corresponding to pairings in which the invader outcompetes the resident and black zones corresponding to pairings in which the resident outcompetes the invader, leading to the latter's extinction. Here, the strategy is the intrinsic tendency to sprout, given as a scalar deviation on the *y*-intercept of the linear model of sprouting used to parameterize matrix projection matrices. (a) An invader with greater sprouting tendency always outcompetes a resident with lower sprouting tendency, suggesting that perfect sprouting should evolve and that vegetative dormancy is maladaptive. (b) The observed level of sprouting is an evolutionary stable strategy, since invading strategies outcompete residents when the former sprout more than the latter but both sprout less than the observed population and when the former sprout less than the latter but both sprout more than the observed population. (c) An evolutionary stable sprouting strategy exists that includes vegetative dormancy, as in (b), but the predicted ESS is significantly different from the observed sprouting tendency, which is given as the midpoint of the graph (intersection of the 0 deviation lines on both the *x*- and *y*-axes).

significantly from the observed sprouting level ($d = 0$ in eqn 1). We considered our best-fit invasion model to be the simplest model that predicted equilibrium sprouting frequencies that did not differ significantly from those observed (Fig. 2b). The hypothesis that vegetative dormancy is adaptive in response to a generally harsh environment and/or short-term reproductive costs would be supported if the simplest such model was the deterministic, ahistorical model. The hypothesis that vegetative dormancy is a bet-hedging trait would be supported if the simplest such model was the stochastic, ahistorical model. Finally, the hypothesis that vegetative dormancy is an adaptive response to long-term trade-offs such as growth costs, potentially operating within a generally harsh environment, would be supported if the simplest such model was the deterministic, historical model. All analyses were conducted in R 3.0.2 (R Core Team 2013). Further technical details on are provided in Appendix S1.

Results

Population dynamics and characteristics

Both populations remained relatively stable in size during the study. The *Anemone* population ranged in observable size from 1604 individuals in 2000 to 2141 individuals in 2002 (range of annual λ: 0.853–1.284). The *Cypripedium* population ranged in observable size from 381 individuals in 1996 to a peak of 574 individuals in 1999, which was followed by a slow decline to 304 individuals in 2011 (range of annual λ: 0.879–1.215). The density of individuals ranged from 1 to 29 per m^2 in *Anemone* (10.586 ± 0.081; mean ± 1 SE) and from 1 to 45 per m^2 in *Cypripedium* (7.016 ± 0.120; mean ± 1 SE) with each *Cypripedium* individual composed of 1.093 ± 0.013 sprouts on average. The minimum fraction of the population that was dormant per year was 0.133 ± 0.030 in *Anemone* (mean ± 1 SE; range: 0.071 in 2002 to 0.217 in 2004) and 0.321 ± 0.029 in *Cypripedium* (range: 0.121 in 1996 to 0.536 in 2008). The longest consecutive number of years in which a plant was observed to be dormant was 15 and 6 years in *Cypripedium* and *Anemone*, respectively.

Density dependence

We identified negative density dependence in the probabilities of sprouting and flowering for *Cypripedium* and only for the probability of flowering in *Anemone*. The mixed model of sprouting in *Cypripedium* suggested a steeper decline in sprouting probability for adult *Cypripedium* plants than for juveniles (coefficient for density in adult model: −0.024 ± 0.003, $P < 0.0001$; juvenile model: −0.018 ± 0.007, $P = 0.0005$; Fig. 3). The mixed models for flowering probability also suggested steep declines with increasing density (coefficient for density in *Cypripedium* model: −0.049 ± 0.005, $P < 0.0001$; *Anemone* model: −0.021 ± 0.001, $P < 0.0001$, Fig. 3). Survival was positively density-dependent in adult and juvenile *Cypripedium*, and in juvenile *Anemone*, suggesting that survival varies spatially with environmental quality (coefficient for density in adult *Cypripedium* model: 0.219 ± 0.019, $P < 0.0001$; juvenile *Cypripedium* model: 0.177 ± 0.040, $P < 0.0001$; adult *Anemone* model: 0.0004 ± 0.0008, $P = 0.638$; juvenile *Anemone* model: 0.005 ± 0.002, $P = 0.010$).

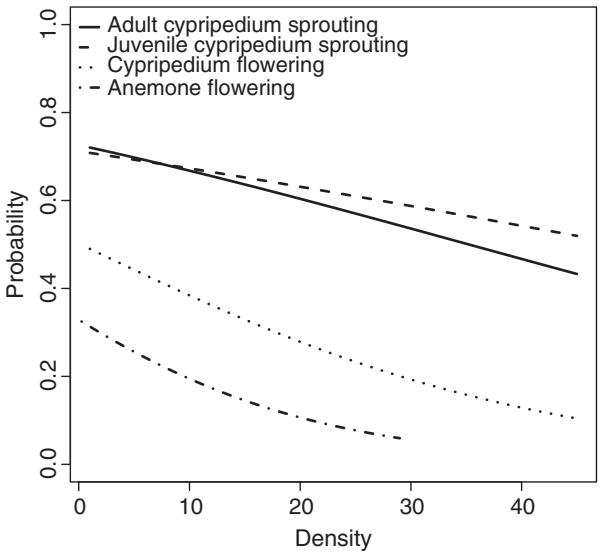

Fig. 3. Negative density dependence in key vital rates of *Anemone* and *Cypripedium*. Adult and juvenile sprouting probability in year $t + 1$, conditional upon surviving, in *Cypripedium*, and flowering probability in year t in adult *Cypripedium* and adult *Anemone*. Density is the number of physiologically integrated individuals per 1 m^2.

Evolutionary dynamics

Game theoretical analyses suggested that observed sprouting rates involving vegetative dormancy are adaptive primarily due to long-term life-history trade-offs. Deterministic ahistorical, stochastic ahistorical and deterministic historical models yielded optimally intermediate sprouting rates in *Cypripedium*, suggesting that short- and long-term life-history trade-offs, a generally harsh environment and temporal environmental stochasticity all likely contribute to the evolution of vegetative dormancy (Fig. 4a–c). However, only the historical deterministic model yielded sprouting frequencies not significantly different from those observed in the population (optimal deviation on y-intercept of sprouting in historical deterministic model for *Cypripedium* assuming density-dependent sprouting: $\beta_{0,spr} = 0.508 \pm 0.441$; mean ± 1 SE; Fig. 5a–e). In *Anemone*, only the historical deterministic model yielded ESS sprouting frequencies below unity (Fig. 4d–f), and these were not significantly different from observed sprouting frequencies (optimal deviation on y-intercept of sprouting in historical deterministic model for *Anemone* assuming density-dependent flowering: $\beta_{0,spr} = 0.379 \pm 0.545$; Fig. 5f–j). Thus, long-term costs of growth and reproduction strongly contribute to the evolution of vegetative dormancy in *Anemone*.

Life-history costs

The cost of growth to survival favoured vegetative dormancy in both species. Our best-fit ahistorical mixed models

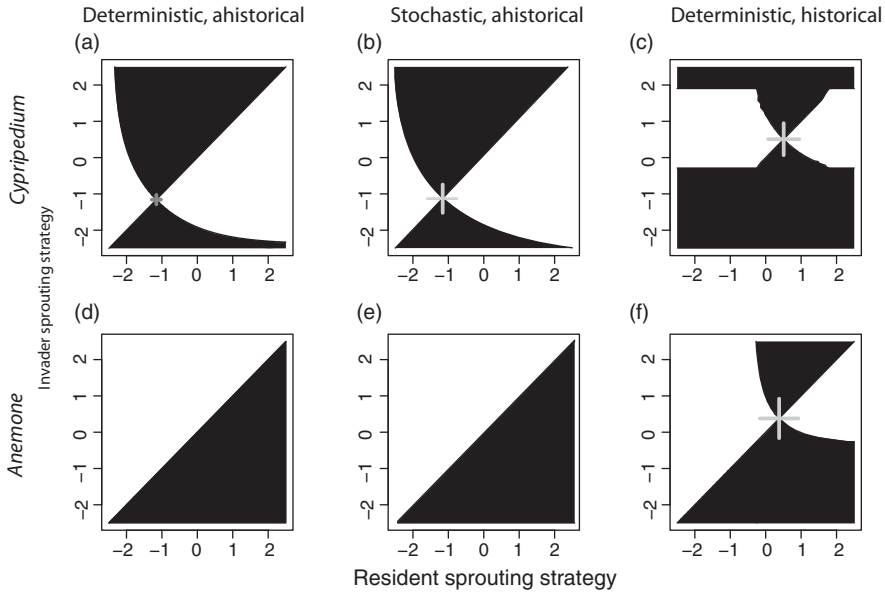

Fig. 4. Pairwise invasibility plots of sprouting strategies show convergence stable ESS sprouting in response to long-term costs of growth in *Cypripedium* (a–c) and *Anemone* (d–f). (a, d) Ahistorical deterministic models yield intermediate sprouting as an ESS in *Cypripedium* but not *Anemone*, but do not overlap with observed sprouting levels. (b, e) Ahistorical stochastic models yield intermediate sprouting as an ESS in *Cypripedium* but not *Anemone*, but do not overlap with observed sprouting levels. (c, f) Historical deterministic models converge on intermediate sprouting as an ESS and overlap with observed sprouting levels in both species. *x*- and *y*-axes indicate altered intrinsic sprouting tendency, given as a deviation to the *y*-intercept in the best-fit general linear mixed model of sprouting in the resident versus invading strategy, respectively. Black and white regions indicate pairs of strategies leading to extinction and maintenance of the invader, respectively. Grey crosshairs indicate stable convergence ESS ± 1SE.

supported a cost of greater size in year t on survival to year $t + 1$ in juvenile plants of both species and in adult *Cypripedium* plants (Table S1), while historical models were consistent with a cost of growth between years t and $t-1$ on both juvenile and adult survival (Table S2). These trade-offs led to a greater risk of mortality in particular for juvenile plants with high growth from year $t-1$ to year t (Fig. 6a,d). Non-flowering adult *Cypripedium* plants also experienced these costs, with plants that maintain a high size for 2 years in a row having extremely high mortality in the third year (Fig. 6b). Historical models also suggested that flowering in years t and $t-1$ led to increased mortality in *Cypripedium* (Table S2). A significant negative interaction between size and flowering status in year t in models of flowering probability in year $t + 1$ suggested that large flowering plants were less likely to flower in the next year, while small, non-flowering plants were more likely to flower (Tables S1 and S2). Additionally, historical models suggested a negative impact of growth between years $t-1$ and t on sprouting probability in adult *Cypripedium*, flowering probability in *Anemone* and size in year $t + 1$ in both species (Table S2).

Climate and vital rates

All of *Cypripedium*'s vital rates were influenced by climatic variation. The best-fit climatic model of adult survival suggested that survival covaried positively with mean annual temperature, number of freezing days and total annual precipitation, while the best-fit climatic model of juvenile survival suggested a small negative influence of mean annual temperature and a small, positive influence of number of freezing days (Table S3). Juvenile and adult sprouting were both positively influenced by number of freezing days and total annual precipitation (Table S3). The probability of flowering increased with mean annual temperature and number of freezing days, although number of freezing days negatively influenced the number of flowers produced (Table S3). The probability of fruiting conditional on flowering was negatively related to all three climatic variables (Table S3), and the number of fruits conditional on fruiting was negatively related to number of freezing days (Table S3).

Dormancy in juveniles versus adults

Optimal sprouting frequencies included high levels of vegetative dormancy primarily because of the cost of above-ground growth to juvenile survival. While dormant plants were predicted to have higher survival than small sprouting plants, larger flowering adults were predicted to have even higher survival while larger juveniles were predicted to have lower survival (Fig. 6). However, in adult *Anemone*, flowering also led to sufficiently strong reductions in survival to drive the optimal sprouting frequency lower with increasing size (Fig. 5j). Invasion analyses in which we altered sprouting strategies either for juveniles or for adults only yielded intermediate ESS sprouting levels in juvenile, but not adult, *Cypripedium* (Fig. S3a, b). In *Anemone*, altering sprouting strategies separately in juveniles or in adults consistently yielded ESS sprouting strategies in which no vegetative dormancy was favoured (Fig. S3c, d), suggesting that long-term

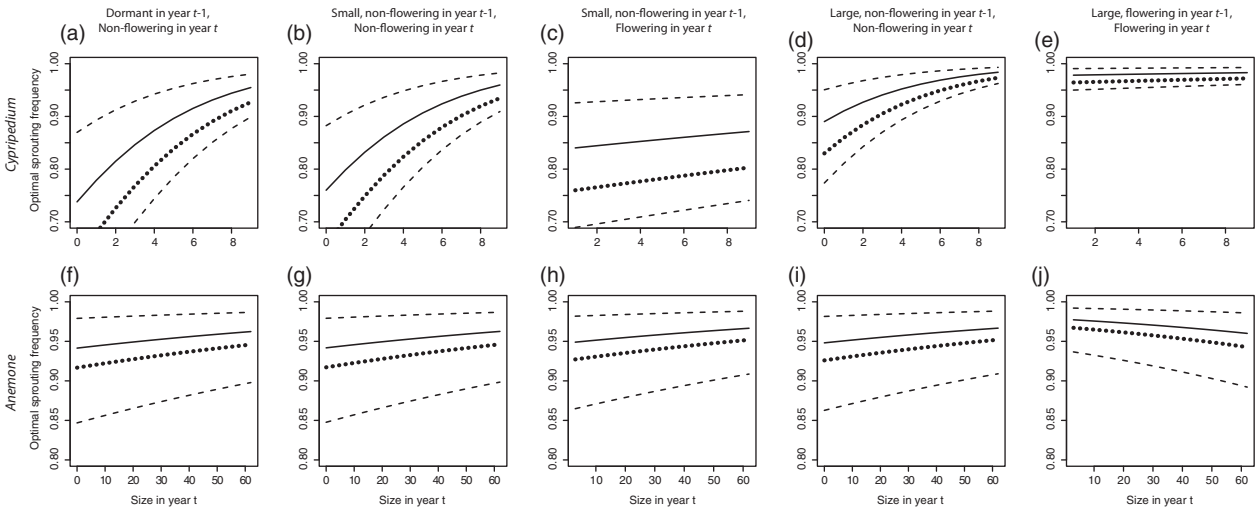

Fig 5. Optimal sprouting frequency is consistently less than 1000 for all sizes of adult plants, in *Cypripedium* (a–e) and *Anemone* (f–j). Sprouting frequency was predicted from the best-fit general linear mixed models (Table S2) of sprouting contingent on survival for plants that were dormant at time $t-1$ and non-flowering at time t (a, f), small and non-flowering at time $t-1$ and non-flowering at time t (b, g), small and non-flowering at time $t-1$ and flowering at time t (c, h), large and non-flowering at time $t-1$ and non-flowering at time t (d, i), and large and flowering at time $t-1$ and flowering at time t (e, j). Black solid lines indicate predicted sprouting frequencies at ESS. Grey dotted lines indicate observed sprouting frequencies. Dashed black lines indicate 95% confidence intervals around ESS sprouting frequencies.

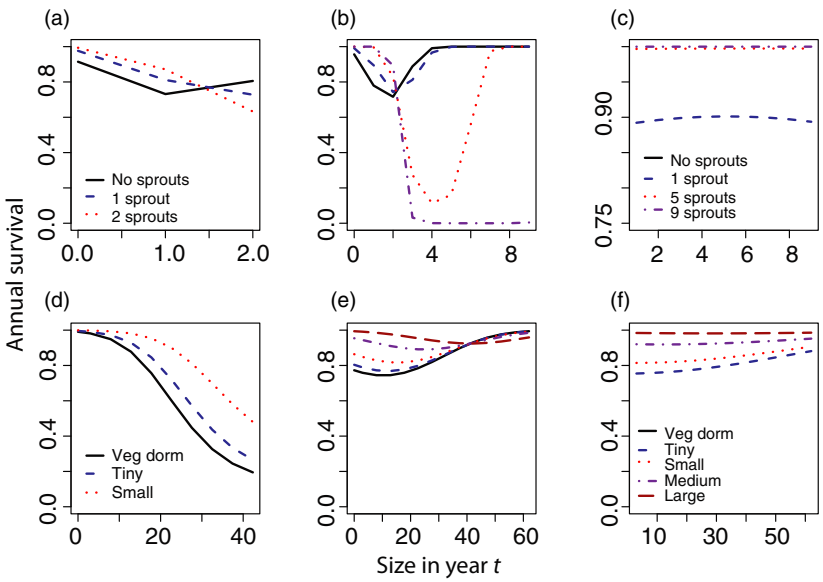

Fig. 6. Juvenile and adult survival depend on individual history in *Cypripedium* (a–c) and *Anemone* (d–f). Larger juveniles have lower survival than smaller juveniles and large, non-flowering adults maintain high survival only by occasionally transitioning to and from small size or vegetative dormancy. Survival was predicted from the best-fit general linear mixed model of survival for each species (Table S2). Survival probability from year t to $t + 1$ of juveniles (a, d), non-flowering adults (b, e) and flowering adults (c, f). Size in year t is indicated on the x-axis, while plotted lines indicate size at time $t-1$ (legends in panels a and d apply only to those panels, while legend in panels c and f also apply to panels b and e). Size categories of *Anemone* are only examples and relate to the length of the basal leaf (tiny: 3 cm, small: 10 cm, medium: 29 cm, and large: 62 cm).

growth costs and environmental pressures operating across the life span make vegetative dormancy adaptive.

Discussion

Natural sprouting frequencies in *Anemone* and *Cypripedium* matched those expected for an ESS driven primarily by life-history costs, particularly the long-term costs of growth. Rapid growth increased mortality in the longer term, particularly for juveniles, which are likely to be small, to lack the resource reserves of adult plants, and to have initially high but declining mortality as they age and grow to adulthood (Harper 1977; Bierzychudek 1982). We expect that such growth costs in juveniles may often be observable even in ahistorical matrix models, because large juveniles only become large due to high, quick growth. Moreover, this

demographic cost is sufficiently strong in juvenile *Cypripedium* that it can explain naturally imperfect sprouting frequencies in that species alone, without consideration of adult growth costs, even in ahistorical analyses. In adults, large plants have lower levels of vegetative dormancy and mortality than small plants, unless they grew large quickly from small size or maintained large size over several years. Unmitigated, such high growth increases mortality risk and so decreases the overall chance of flowering in the future. The combination of these costs in juvenile and adult stages of life favoured vegetative dormancy in *Anemone*.

The conditions that have favoured a long life span in many herbaceous perennials may also explain why vegetative dormancy is adaptive. Iteroparous life histories, involving a long juvenile period and a long reproductive life span, are often adaptations to high extrinsic mortality in juveniles, particularly when mortality is strongly stochastic through time (Metcalfe & Monaghan 2003; Wilbur & Rudolf 2006). Naturally, quick growth in juveniles should be favoured as it leads to the potential for earlier reproduction and greater fecundity. However, in a generally harsh but temporally variable environment, a relatively good year that favoured high growth may be followed by a relatively harsh year in which limited resources, harsh conditions or other barriers prevent the plant from recuperating from such high growth (Shefferson & Roach 2010). Shifting environmental conditions may thus cause resource limitation and other types of stress immediately following a growth spurt and may exacerbate the impact of some life-history trade-offs (Bell & Kofopanou 1986). These extrinsic conditions may include climatic conditions, which could drive physiological response directly or can indirectly influence the plant via other members of the community, and may also include shifting levels of intra- and interspecific competition across the population. Within this context, vegetative dormancy may ease what would otherwise be heightened juvenile mortality following high growth by providing greater flexibility in response to shifting environmental conditions. Thus, vegetative dormancy may best be viewed as an adaptive strategy that allows plants to minimize the mortality-exacerbating impact of above-ground growth under harsh and unpredictable environmental conditions.

Temporal environmental stochasticity has favoured the evolution of many important life-history traits, such as flowering time in long-lived monocarps (Metcalf *et al.* 2008) and seed dormancy (Rees 1996). Vegetative dormancy in *Cypripedium* particularly, but other species as well, may be indirectly tied to temporal environmental stochasticity if it ameliorates harsh periods by preventing above-ground growth from contributing to mortality during sensitive stages of the plant's life. While our results are consistent with this hypothesis, they do not suggest that vegetative dormancy functions as a bet-hedging trait in these two species because such an explanation requires that it be adaptive directly in response to temporal stochasticity and that its effect is to reduce short-term fitness but maximize long-term fitness via a meaningful reduction in the temporal variance in fitness (Seger & Brockman 1987). Our results suggest that this was not the case, particularly in *Anemone*, where the ahistorical stochastic model did not yield optimally intermediate sprouting levels (Fig. 4e).

What stressful conditions may favour vegetative dormancy? Negatively density-dependent sprouting in *Cypripedium* and flowering in *Anemone* suggest that intraspecific competition may create stressful conditions that are ameliorated by imperfect sprouting frequencies. Shifts in presence, abundance and density of dormancy-prone species across gradients of soil moisture, light availability, and density of conspecifics and other species suggest that precipitation, microsite hydrology, local canopy density and competition may all moderate stress levels in ways relevant to the determination of optimal sprouting frequencies (Diez & Pulliam 2007; Hutchings 2010). Interspecific competition may also play a role, particularly if population density exhibits a strong spatial correlation across species. Temporal correlations between sprouting frequency and climatic variables such as annual temperature and precipitation reinforce the importance of the environment in setting stress levels (Shefferson *et al.* 2001; Kéry & Gregg 2004; Hutchings 2010).

Does the possibility still exist that vegetative dormancy is not adaptive? We believe that our study solidly supports the hypothesis that vegetative dormancy is generally adaptive, but there are important caveats. First, evolutionary inferences can depend on fitness metric and means of analysis. Our findings contrast with the results of many matrix analyses of herbaceous perennial demography, which have sometimes suggested that vegetative dormancy may be maladaptive because of its association with higher mortality in the year of dormancy (Hutchings 1987; Shefferson *et al.* 2003; Gregg & Kery 2006). At other times, these analyses have suggested that vegetative dormancy is non-adaptive because it has a minor influence on the asymptotic population growth rate (Salguero-Gómez & Casper 2010). In the former case, fitness is assumed to covary linearly with survival and in both cases fitness is assumed to be density- and frequency-independent. We argue that these are unrealistic assumptions. Our analyses assume that fitness is some function of mortality and fecundity that may be density- and frequency- dependent. Nonetheless, other fitness metrics that rely less on these assumptions may yield different insights.

Second, our inferences may depend on the life-history context of the species chosen for study. Since we used two long-lived perennial species in this study, we do not know whether the adaptive dimensions of vegetative dormancy change as a function of longevity or reproductive life span. Since population growth rate and fitness are more sensitive to fecundity than to survival in short-lived than in long-lived species (Sæther & Bakke 2000), vegetative dormancy may not be commonly favoured among such species. Similarly, longer-term growth costs may also be important in at least some long-lived species. Given the computationally intensive nature of the Ehrlen approach, particularly when high-dimension matrices such as IPMs are used, individual-based modelling should be used to assess the influence of longer-term life-history costs.

Third, some cases of vegetative dormancy may have been misidentified, being instead cases where above-ground parts

have been lost through herbivory, or cases of sprouting date varying among individuals (Gilbert & Lee 1980; Mehrhoff 1989; Gregg 2011). However, we discount this possibility as a major issue for our study systems because if such cases were important in our study, we should not have obtained results suggesting that imperfect sprouting is adaptive. Further, strong herbivory would require adaptive mechanisms for survival in the face of this stress, potentially for many years without photosynthesis. Therefore, what we term vegetative dormancy would be a potentially adaptive tolerance mechanism to promote survival in a defoliated state.

In conclusion, we have presented strong support for the proposal that vegetative dormancy is an adaptive trait in herbaceous perennial plant species. It is adaptive because the environment is likely to be generally harsh but variable, because growth can be costly in the short and/or long term and because vegetative dormancy provides a means for reduction or postponement of this cost. Our results imply a strong difference in physiological cost between producing/maintaining above-ground sprouts and maintaining a rootstock without such structures.

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Data Accessibility

Matrices used in these analyses have been deposited in the Dryad Digital Repository (Shefferson *et al.* 2014).

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Sample ahistorical matrix model for *Cypripedium* with relevant vital rates identified for each transition.

Figure S2. Sample historical matrix model for *Cypripedium* with relevant vital rates identified for each transition.

Figure S3. Tests of whether ESS sprouting levels predicting vegetative dormancy are due to costs operating in juveniles, adults or both, in *Cypripedium* and *Anemone*.

Table S1. Best-fit linear mixed models of vital rates in ahistorical models of *Cypripedium* and *Anemone*.

Table S2. Best-fit linear mixed models of vital rates in historical models of *Cypripedium* and *Anemone*.

Table S3. Tests of the influence of climate on vital rates in *Cypripedium*.

Appendix S1. Supplemental invasion analysis methods.